

Advanced Calculus HW 4

Robert O'Connell

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Question 1

(a)

We have

$$\vec{B}(t) = \frac{\vec{r}'(t) \times \vec{r}''(t)}{|\vec{r}'(t) \times \vec{r}''(t)|}$$

Differentiating we get

$$\frac{d}{dt}(\vec{r}' \times \vec{r}'') = \vec{r}'' \times \vec{r}'' + \vec{r}' \times \vec{r}''' = \vec{r}' \times \vec{r}''',$$

and $\frac{d}{dt}|\vec{u}| = \frac{\vec{u} \cdot \vec{u}'}{|\vec{u}|}$, then we get

$$\begin{aligned} \vec{B}' &= \frac{\vec{r}' \times \vec{r}'''}{|\vec{r}' \times \vec{r}''|} - \frac{(\vec{r}' \times \vec{r}'')}{|\vec{r}' \times \vec{r}''|^2} \cdot \frac{(\vec{r}' \times \vec{r}'') \cdot (\vec{r}' \times \vec{r}''')}{|\vec{r}' \times \vec{r}''|} \\ &= \frac{\vec{r}' \times \vec{r}'''}{|\vec{r}' \times \vec{r}''|} - \frac{[(\vec{r}' \times \vec{r}'') \cdot (\vec{r}' \times \vec{r}''')](\vec{r}' \times \vec{r}'')}{|\vec{r}' \times \vec{r}''|^3}. \end{aligned}$$

Calculating $\vec{N}$ using $\vec{N} = \vec{B} \times \vec{T}$,

$$\vec{N} = \frac{\vec{r}' \times \vec{r}''}{|\vec{r}' \times \vec{r}''|} \times \frac{\vec{r}'}{|\vec{r}'|} = \frac{(\vec{r}' \times \vec{r}'') \times \vec{r}'}{|\vec{r}'||\vec{r}' \times \vec{r}''|}.$$

Expanding using property $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$:

$$\begin{aligned} (\vec{r}' \times \vec{r}'') \times \vec{r}' &= (\vec{r}' \cdot \vec{r}')\vec{r}'' - (\vec{r}'' \cdot \vec{r}')\vec{r}' = |\vec{r}'|^2\vec{r}'' - (\vec{r}' \cdot \vec{r}'')\vec{r}', \\ \vec{N} &= \frac{|\vec{r}'|^2\vec{r}'' - (\vec{r}' \cdot \vec{r}'')\vec{r}'}{|\vec{r}'||\vec{r}' \times \vec{r}''|}. \end{aligned}$$

Now we compute $\vec{B}' \cdot \vec{N}$. The second term of $\vec{B}'$ involves $(\vec{r}' \times \vec{r}'') \cdot \vec{N}$, which vanishes since $\vec{r}' \times \vec{r}''$ is parallel to $\vec{B}$ and $\vec{B}$ is perp to $\vec{N}$. Therefore

$$\begin{aligned} \vec{B}' \cdot \vec{N} &= \frac{\vec{r}' \times \vec{r}'''}{|\vec{r}' \times \vec{r}''|} \cdot \frac{|\vec{r}'|^2\vec{r}'' - (\vec{r}' \cdot \vec{r}'')\vec{r}'}{|\vec{r}'||\vec{r}' \times \vec{r}''|} \\ &= \frac{1}{|\vec{r}'||\vec{r}' \times \vec{r}''|^2} \left[(\vec{r}' \times \vec{r}''') \cdot (|\vec{r}'|^2\vec{r}'' - (\vec{r}' \cdot \vec{r}'')\vec{r}') \right]. \end{aligned}$$

Since $(\vec{r}' \times \vec{r}''') \cdot \vec{r}' = 0$ (perp) and $(\vec{r}' \times \vec{r}''') \cdot \vec{r}'' = (\vec{r}' \times \vec{r}'') \cdot \vec{r}'''$ (property of cross products), we get

$$= -\frac{|\vec{r}'|^2(\vec{r}' \times \vec{r}''') \cdot \vec{r}''}{|\vec{r}'||\vec{r}' \times \vec{r}''|^2} = -\frac{|\vec{r}'|(\vec{r}' \times \vec{r}'') \cdot \vec{r}'''}{|\vec{r}' \times \vec{r}''|^2}.$$

We have, $\vec{B}' = -\tau|\vec{r}'|\vec{N}$, dotting both sides with the unit vector $\vec{N}$ gives $\vec{B}' \cdot \vec{N} = -\tau|\vec{r}'|$.

$$\begin{aligned} \Rightarrow \tau|\vec{r}'| &= \frac{|\vec{r}'|(\vec{r}' \times \vec{r}'') \cdot \vec{r}'''}{|\vec{r}' \times \vec{r}''|^2} \\ \Rightarrow \tau &= \frac{(\vec{r}' \times \vec{r}'') \cdot \vec{r}'''}{|\vec{r}' \times \vec{r}''|^2}. \end{aligned}$$

(b)

First show $\vec{B}$ is constant, i.e. $\vec{B}' = 0$

From we have (a),

$$\vec{B}' = \frac{\vec{r}' \times \vec{r}'''}{|\vec{r}' \times \vec{r}''|} - \frac{[(\vec{r}' \times \vec{r}'') \cdot (\vec{r}' \times \vec{r}''')](\vec{r}' \times \vec{r}'')}{|\vec{r}' \times \vec{r}''|^3}.$$

We have $\vec{r}'' = \alpha(t)\vec{r}$, so

$$\vec{r}''' = \alpha'(t)\vec{r} + \alpha(t)\vec{r}'.$$

We compute $\vec{r}' \times \vec{r}'''$

$$\vec{r}' \times \vec{r}''' = \vec{r}' \times (\alpha'(t)\vec{r} + \alpha(t)\vec{r}') = \alpha'(t)(\vec{r}' \times \vec{r}) + \alpha(t)(\vec{r}' \times \vec{r}') = \alpha'(t)(\vec{r}' \times \vec{r}),$$

and $\vec{r}' \times \vec{r}''$:

$$\vec{r}' \times \vec{r}'' = \vec{r}' \times \alpha(t)\vec{r} = \alpha(t)(\vec{r}' \times \vec{r}).$$

Therefore $\vec{r}' \times \vec{r}'''$ and $\vec{r}' \times \vec{r}''$ are both parallel to $\vec{r}' \times \vec{r}$,

with $\vec{w} = \vec{r}' \times \vec{r}$ we have

$$\vec{r}' \times \vec{r}'' = \alpha(t)\vec{w}, \quad \vec{r}' \times \vec{r}''' = \alpha'(t)\vec{w}.$$

Then

$$\begin{aligned} \vec{B}' &= \frac{\alpha'(t)\vec{w}}{|\alpha(t)\vec{w}|} - \frac{(\alpha(t)\vec{w} \cdot \alpha'(t)\vec{w})\alpha(t)\vec{w}}{|\alpha(t)\vec{w}|^3} = \frac{\alpha'(t)\vec{w}}{|\alpha(t)||\vec{w}|} - \frac{\alpha(t)\alpha'(t)|\vec{w}|^2 \cdot \alpha(t)\vec{w}}{|\alpha(t)|^3|\vec{w}|^3} \\ &= \frac{\alpha'(t)\vec{w}}{|\alpha(t)||\vec{w}|} - \frac{\alpha'(t)\vec{w}}{|\alpha(t)||\vec{w}|} = \vec{0}. \end{aligned}$$

Hence $\vec{B}$ is constant

Now compute $\tau(t)$

$$\tau = \frac{(\vec{r}' \times \vec{r}'') \cdot \vec{r}'''}{|\vec{r}' \times \vec{r}''|^2}.$$

We have $\vec{r}' \times \vec{r}'' = \alpha(t)\vec{w}$ and $\vec{r}''' = \alpha'(t)\vec{r} + \alpha(t)\vec{r}'$, so

$$(\vec{r}' \times \vec{r}'') \cdot \vec{r}''' = \alpha(t)\vec{w} \cdot (\alpha'(t)\vec{r} + \alpha(t)\vec{r}') = \alpha(t)\alpha'(t)(\vec{r}' \times \vec{r}) \cdot \vec{r} + \alpha(t)^2(\vec{r}' \times \vec{r}) \cdot \vec{r}'.$$

Both terms vanish, since we are dotting a vector with a vector defined to be perpendicular to it
Hence

$$(\vec{r}' \times \vec{r}'') \cdot \vec{r}''' = 0,$$

Hence

$$\tau(t) = 0.$$

Question 2

$$\vec{r}(t) = (1 - t^2)\hat{i} - t^2\hat{j}$$

$$\vec{v}(t) = \vec{r}'(t) = -2t\hat{i} - 2t\hat{j}$$

Speed is the magnitude of $\vec{v}(t)$

$$|\vec{v}(t)| = \sqrt{8t^2}$$

The minimum speed of the particle is when $t = 0$, $|\vec{v}(t)| = 0$, since $\sqrt{x} \geq 0 \forall x$

The corresponding location is $\vec{r}(0)$

$$\vec{r}(0) = \hat{i}$$

Question 3

(a)

$$\vec{r}(t) = (t^2 - 2t)\hat{i} + (t^2 - 4)\hat{j}$$

$$\vec{v}(t) = (2t - 2)\hat{i} + 2t\hat{j}$$

$$\vec{a}(t) = 2\hat{i} + 2\hat{j}$$

$$|\vec{v}(t)| = 2\sqrt{2t^2 - 2t + 1}$$

$$\vec{v}(t) \cdot \vec{a}(t) = 8t - 4$$

$$\vec{v}(t) \times \vec{a}(t) = ((4t - 4) - 4t)\hat{k} = 4\hat{k}$$

$$|\vec{v}(t) \times \vec{a}(t)| = 4$$

$$\vec{T}(t) = \frac{(2t - 2)\hat{i} + 2t\hat{j}}{2\sqrt{2t^2 - 2t + 1}}$$

In arbitrary parameterizations

$$\vec{N}(t) = \frac{(\vec{v}(t) \times \vec{a}(t)) \times \vec{v}(t)}{|(\vec{v}(t) \times \vec{a}(t)) \times \vec{v}(t)|}$$

$$(\vec{v}(t) \times \vec{a}(t)) \times \vec{v}(t) = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 4 \\ 2t - 2 & 2t & 0 \end{bmatrix} = (-8t)\hat{i} + (8t - 8)\hat{j}$$

$$|(\vec{v}(t) \times \vec{a}(t)) \times \vec{v}(t)| = \sqrt{128t^2 - 128t + 64}$$

$$\vec{N}(t) = \frac{(-8t)\hat{i} + (8t - 8)\hat{j}}{\sqrt{128t^2 - 128t + 64}}$$

Then,

$$a_T = \frac{8t - 4}{2\sqrt{2t^2 - 2t + 1}} = \frac{4t - 2}{\sqrt{2t^2 - 2t + 1}}$$

$$a_N = \frac{4}{2\sqrt{2t^2 - 2t + 1}} = \frac{2}{\sqrt{2t^2 - 2t + 1}}$$

$$\vec{a}(t) = \frac{4t - 2}{\sqrt{2t^2 - 2t + 1}} \frac{(2t - 2)\hat{i} + 2t\hat{j}}{2\sqrt{2t^2 - 2t + 1}} + \frac{2}{\sqrt{2t^2 - 2t + 1}} \frac{(-8t)\hat{i} + (8t - 8)\hat{j}}{\sqrt{128t^2 - 128t + 64}}$$

$$\kappa(t) = \frac{4}{(2\sqrt{2t^2 - 2t + 1})^3} = \frac{1}{2(2t^2 - 2t + 1)^{\frac{3}{2}}}$$

$$\kappa(1) = \frac{1}{2}$$

(b)

$$\vec{r}(t) = 3 \sin t \hat{i} + 2 \cos t \hat{j} - \sin 2t \hat{k}$$

$$\vec{v}(t) = 3 \cos t \hat{i} - 2 \sin t \hat{j} - 2 \cos 2t \hat{k}$$

$$\vec{a}(t) = -3 \sin t \hat{i} - 2 \cos t \hat{j} + 4 \sin 2t \hat{k}$$

$$|\vec{v}(t)| = \sqrt{9 \cos^2 t + 4 \sin^2 t + 4 \cos^2 2t}$$

$$\vec{v}(t) \cdot \vec{a}(t) = -9 \cos t \sin t + 4 \sin t \cos t - 8 \sin 2t \cos 2t$$

$$\begin{aligned} \vec{v}(t) \times \vec{a}(t) &= (-8 \sin t \sin 2t - 4 \cos t \cos 2t)\hat{i} + (12 \cos t \sin 2t - 6 \sin t \cos 2t)\hat{j} + (-6 \cos^2 t - 6 \sin^2 t)\hat{k} \\ &= -4(2 \sin t \sin 2t - \cos t \cos 2t)\hat{i} + 6(2 \cos t \sin 2t - \sin t \cos 2t)\hat{j} + (-6)\hat{k} \end{aligned}$$

$$\begin{aligned}
& |\vec{v}(t) \times \vec{a}(t)|^2 = \\
& 16(\sin^2 t \sin^2 2t - 4 \sin t \cos t \sin 2t \cos 2t + \cos^2 t \cos^2 2t) + 36(4 \cos^2 t \sin^2 2t - 4 \sin t \cos t \sin 2t \cos 2t + \sin^2 t \cos^2 2t) + 36 \\
& \Rightarrow |\vec{v}(t) \times \vec{a}(t)| = \sqrt{616 \cos^2 t \sin^4 t + 164 \cos^4 t \sin^2 t + 16 \cos^6 t + 36 \sin^6 t + 36}
\end{aligned}$$

$$\vec{T}(t) = \frac{3 \cos t \hat{i} - 2 \sin t \hat{j} - 2 \cos 2t \hat{k}}{\sqrt{9 \cos^2 t + 4 \sin^2 t + 4 \cos^2 2t}}$$

Then,

$$\begin{aligned}
a_T &= \frac{-9 \cos t \sin t + 4 \sin t \cos t - 8 \sin 2t \cos 2t}{\sqrt{9 \cos^2 t + 4 \sin^2 t + 4 \sin^2 2t}} \\
a_N &= \frac{\sqrt{616 \cos^2 t \sin^4 t + 164 \cos^4 t \sin^2 t + 16 \cos^6 t + 36 \sin^6 t + 36}}{\sqrt{9 \cos^2 t + 4 \sin^2 t + 4 \sin^2 2t}}
\end{aligned}$$

We know $\vec{a}(t) = a_T \vec{T} + a_N \vec{N}$, we can use this to solve for $\vec{N}(t)$

$$\begin{aligned}
-3 \sin t \hat{i} - 2 \cos t \hat{j} + 4 \sin 2t \hat{k} &= \frac{-9 \cos t \sin t + 4 \sin t \cos t - 8 \sin 2t \cos 2t}{\sqrt{9 \cos^2 t + 4 \sin^2 t + 4 \sin^2 2t}} \\
&\quad + \frac{3 \cos t \hat{i} - 2 \sin t \hat{j} - 2 \cos 2t \hat{k}}{\sqrt{9 \cos^2 t + 4 \sin^2 t + 4 \cos^2 2t}} + \\
&\quad \frac{\sqrt{616 \cos^2 t \sin^4 t + 164 \cos^4 t \sin^2 t + 16 \cos^6 t + 36 \sin^6 t + 36}}{\sqrt{9 \cos^2 t + 4 \sin^2 t + 4 \sin^2 2t}} \vec{N}
\end{aligned}$$

I am not going to solve for $\vec{N} :$ (

Finally,

$$\begin{aligned}
\kappa(t) &= \frac{\sqrt{616 \cos^2 t \sin^4 t + 164 \cos^4 t \sin^2 t + 16 \cos^6 t + 36 \sin^6 t + 36}}{(9 \cos^2 t + 4 \sin^2 t + 4 \cos^2 2t)^{\frac{3}{2}}} \\
\kappa\left(\frac{\pi}{2}\right) &= \frac{3}{8}
\end{aligned}$$